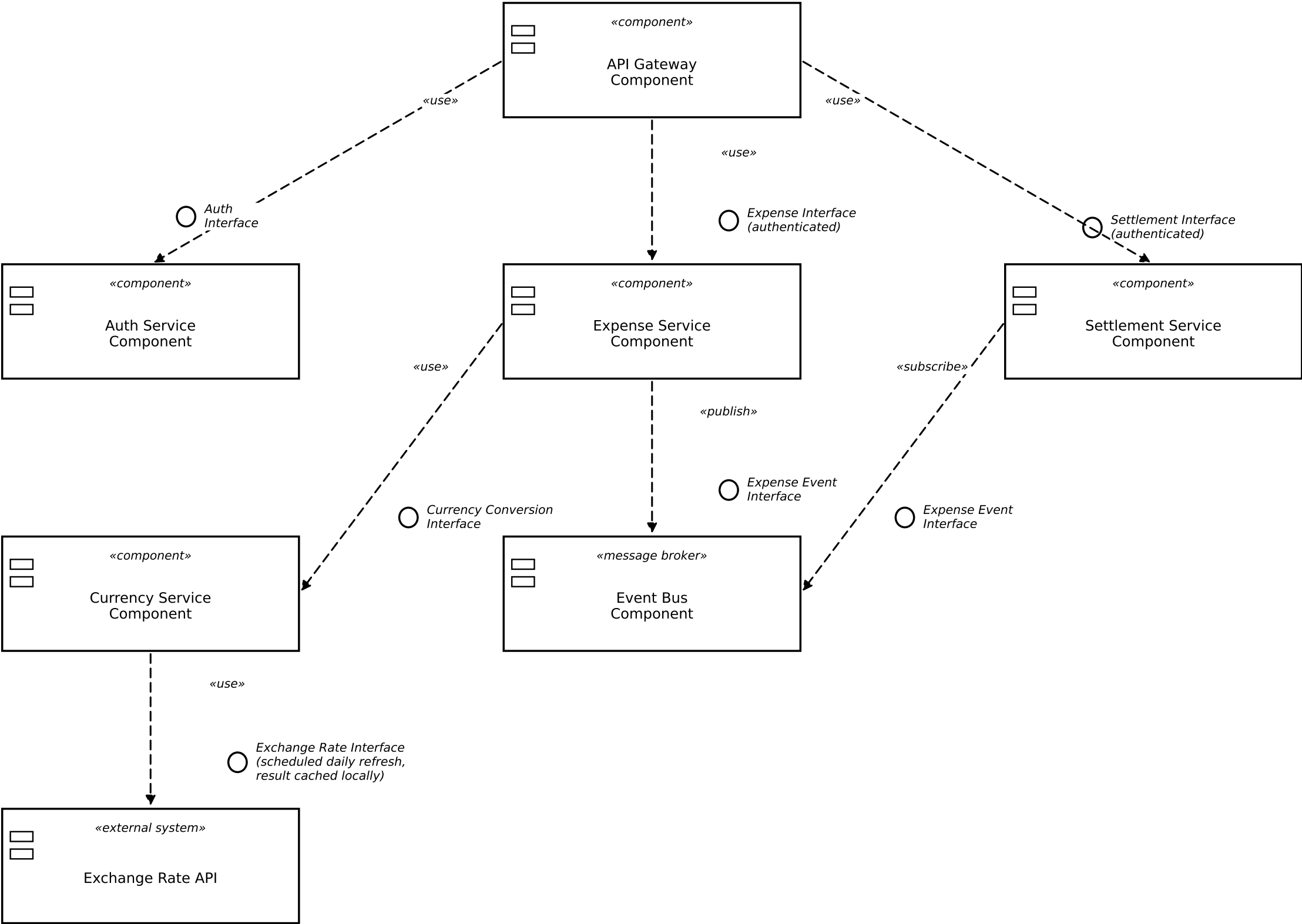


Group Travel Expense Simplification Engine — Microservices Architecture (v2)



Architectural Justification

Group Travel Expense Simplification Engine

We chose Microservices Architecture for the Group Travel Expense Simplification Engine.

Reason 1: The system's core function, computing the minimum-transaction debt settlement graph, is computationally heavier than the rest of the application. To keep this fast and independent, Settlement Service does not call Expense Service directly. Instead, Expense Service publishes expense events to an Event Bus, and Settlement Service subscribes to those events to maintain its own local view of the data. This removes a synchronous cross-service call from the settlement path and lets the Settlement Service be scaled independently without waiting on other services.

Reason 2: Currency conversion depends on an external, third-party exchange-rate feed, which is a dependency outside our control. Since rates only change once per day, Currency Service refreshes and caches the day's rate on a schedule rather than calling the external API on every conversion. This means expense entry and conversion no longer depend on the external API being available or fast at the moment a traveler adds an expense.

Security Advantage: Authentication is centralized in a dedicated Auth Service behind the API Gateway. The Gateway authenticates each request once and attaches the caller's validated role/claims to the request before routing it onward, so Expense Service and Settlement Service can enforce authorization locally from those claims without repeatedly calling Auth Service. This keeps sensitive financial logic from being directly exposed to clients while avoiding unnecessary internal chatter.

Performance Benefit: Decoupling Settlement Service from Expense Service via the Event Bus, and decoupling Currency Service from the external exchange-rate API via local caching, both remove synchronous, blocking calls from the settlement path. This directly supports our performance requirement (NFR-001) that settlement for up to 50 travelers and 500 expenses must complete in under 100 ms, since the Settlement Service's own compute resources can be scaled independently and its critical path no longer waits on other services or external systems.